

Experiment 18

Question

Perform Edge Detection using Sobel Matrix along Y-axis.

Aim

To detect the horizontal edges of an image using the Sobel operator along the Y-axis.

Procedure

1. Import the OpenCV library.
2. Read the image in grayscale mode.
3. Apply the Sobel operator along the Y-axis.
4. Display the original and output images.
5. Save the output image.

Code

```
import cv2

img = cv2.imread("dog.jpg", 0)

sobely = cv2.Sobel(img, cv2.CV_64F, 0, 1, ksize=3)

cv2.imshow("Original", img)
cv2.imshow("Sobel Y", sobely)

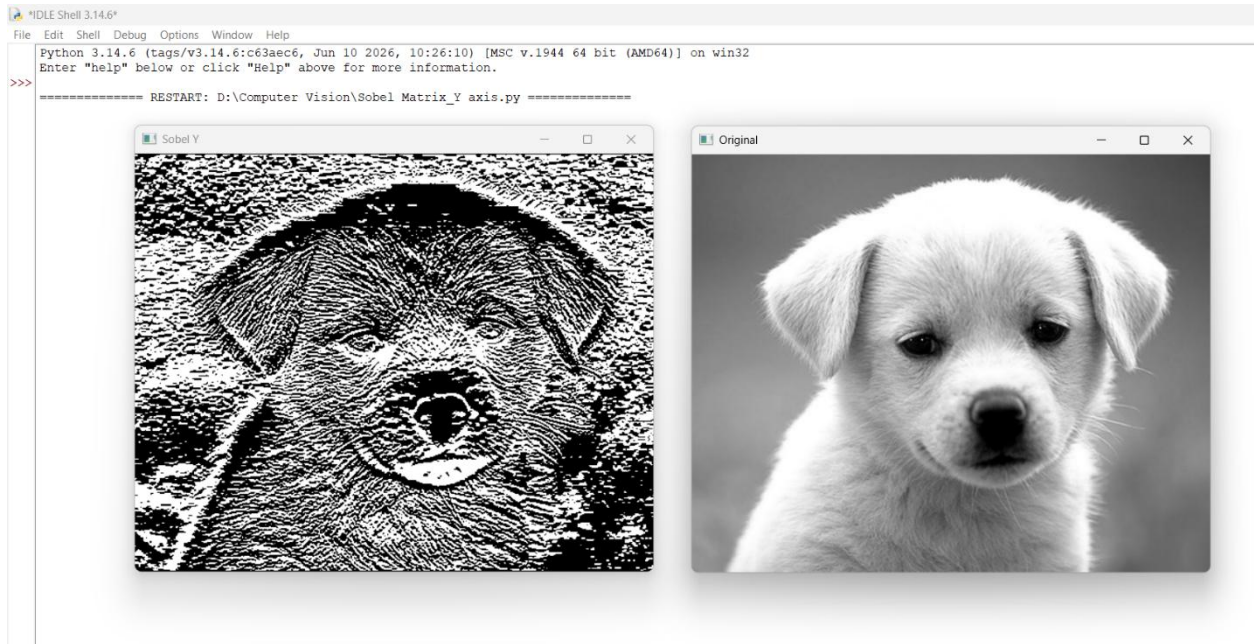
cv2.imwrite("sobely_dog.jpg", sobely)
```

Input

Input Image: dog.jpg

Output

- Sobel Y Edge Image



Result

The horizontal edges of the image were successfully detected using the Sobel operator along the Y-axis.